27 Use of matrices in multiple reactions problems

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Introduction: A major concept in chemical reaction engineering is stoichiometry and this is amplified when dealing with multiple reactions. While the Fogler textbook addresses stoichiometry using the equation of relative rates, i.e., for the reaction aA + bB --> cC + dD, $\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d}$, the book by Rawlings and Ekerdt uses matrices to deal with stiochiometry. In fact, use of matrices to solve chemical reaction engineering problems is quite common, and we will see examples employing matrices when we study reaction mechanisms, as well. In this lecture, I will show how to

1. Write chemical reactions in matrix form, i.e., writing the "stoichiometry matrix"
2. Use the transpose of the "stoichiometry matrix" to write net rates for systems of multiple reactions
3. Identify the number of independent reactions from a system of multiple reactions
4. Create a system of reactions to explain an observed product distribution

Topic 1: Writing the "stoichiometry matrix"

Essentially, this is the practice of writing a chemical reaction as a mathematical equation and setting it equal to zero

For example, consider the nitric oxide oxidation reaction,
2NO + O2 --> 2NO2.

Writing as a mathematical equation:

$$2NO + O_2 = 2NO_2$$

Setting it equal to 0:

$$0 = -2NO - O_2 + 2NO_2$$

We can recover this linear equation using matrices, i.e.,

$$r = \begin{bmatrix} -2 & -1 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} NO \\ O_2 \\ NO_2 \end{bmatrix}$$

Where A is the species matrix, ~~which we denote A.~~ In this case, A1 = NO

A2 = O_2 and A3 = NO_2

ν

is the stoichiometry matrix. Note that stoichiometric coefficients of reactants are negative and stoichiometric coefficients of products are positive

The chemical reaction written in matrix form is

$$\nu A = 0$$

For example, the stoichiometry matrix can also be used to compute ΔH^{rxn} by replacing the species matrix with a "heats of formation" matrix:

$$H = \begin{bmatrix} \Delta H^\circ_{f,NO} \\ \Delta H^\circ_{f,O_2} \\ \Delta H^\circ_{f,NO_2} \end{bmatrix}$$

The rate of reaction 1 is defined relative to species A, the rate of reaction 2 is defined relative to species A, and the rate of reaction 3 is defined relative to species C.

The stoichiometry matrix <u>must</u> be set up so that the stoichiometric coefficient on the species that the rate is written relative to is equal to one. Using the example above:

ΔC_P and really any quantity that takes a "produces minus reactants" form can be computed similarly.

Topic 2: Using the transpose of the stoichiometry matrix to write species' net rates

Consider the multiple reactions:

1 CH4 + O2 <----> CO2 + 2H2

2 2H2 + O2 <----> 2H2O

3 2CO + 4H2 <----> 2CH4 + O2

First, determine an "order" for each species in the species matrix. Let's set A1 = CH4, A2 = O2, A3 = CO2, A4 = H2, A5 = H2O, A6 = CO, A7 = CH4.

Since there are 3 reactions, the stoichiometry matrix has 3 rows

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$$① \quad A + 2B \rightarrow C \qquad -r_{1A} = k_{1A} C_A C_B^2$$

$$② \quad \frac{3}{2} C + A \rightarrow \frac{1}{2} D \qquad -r_{2A} = k_{2A} C_A^2 C_C^3$$

$$③ \quad \frac{1}{3} B + C \rightarrow \frac{1}{3} E \qquad -r_{3C} = k_{3C} C_B C_C^3$$

We can then write $\nu^T r =$

$$\begin{array}{cc} & \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \nu = & \begin{bmatrix} -1 & -2 & 1 & 0 & 0 \\ -1 & 0 & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{1}{3} & -1 & 0 & \frac{1}{3} \end{bmatrix} \end{array}$$

$$r^T = \begin{bmatrix} -1 & -1 & 0 \\ -2 & 0 & -\frac{1}{3} \\ 1 & -\frac{3}{2} & -1 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \qquad r = \begin{bmatrix} k_{1A} C_A C_B^2 \\ k_{2A} C_A^2 C_C^3 \\ k_{3C} C_B C_C^3 \end{bmatrix}$$

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To write the stoichiometry matrix, we tabulate the stoichiometric coefficients of these species in each reaction, i.e.,

$$\nu = \begin{array}{c} \begin{array}{ccccccc} CH_4 & O_2 & CO_2 & H_2 & H_2O & CO & CH_4 \end{array} \\ \begin{bmatrix} -1 & -1 & 1 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 & 2 & 0 & 0 \\ 2 & 1 & 0 & -4 & 0 & -2 & 2 \end{bmatrix} \end{array} \begin{array}{l} \\ \leftarrow \text{rxn } 1 \\ \leftarrow \text{rxn } 2 \\ \leftarrow \text{rxn } 3 \end{array}$$

The net rate of each species is then $\nu^T r$, where r are the rates of rxns 1, 2, and 3

An important note: Rates are defined relative to some species. For example:

1. A + 2B --> C $\quad -r_{1A} = k_{1A} C_A C_B^2$
2. 3C + 2A --> D $\quad -r_{2A} = k_{2A} C_A^2 C_C^3$
3. B + 3C --> E $\quad -r_{3C} = k_{3C} C_B C_C^3$